Punc’data
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Punc’data

Tutorial for assignment

Purpose of this tutorial

- This tutorial is given to help you navigate through data importation in Punc’data and assignment of high resolution mass spectrometry data
- Calibration and the use of canvas (to make figures) will not be adressed here

- The data used in this tutorial is available in the source code of Punc’data (https://github.com/WTVoe/puncdata)

Folder : Examples\testdata_cellulosepyrolysis_POSITIVE_raw.csv

This dataset was acquired using an FT-ICR mass spectrometer 7T 2xR (Bruker Daltonics)

Table of contents

1. Method

The assignment method in Punc'data includes both a systematic « De Novo » approach and a mass difference network approach. The former is the default one, while the latter is adapted to complex mixtures (DOM, NOM, oils, polymers…)

A few interesting studies on the subject:

- Kunenkov et al. Anal. Chem. 2009 https://doi.org/10.1021/ac901476u
- Tziotis et al. Eur. J. Mass Spectrom. 2011 https://doi.org/10.1255/ejms.1135
- Kilgour et al. Anal. Chem. 2012 https://doi.org/10.1021/ac301339d

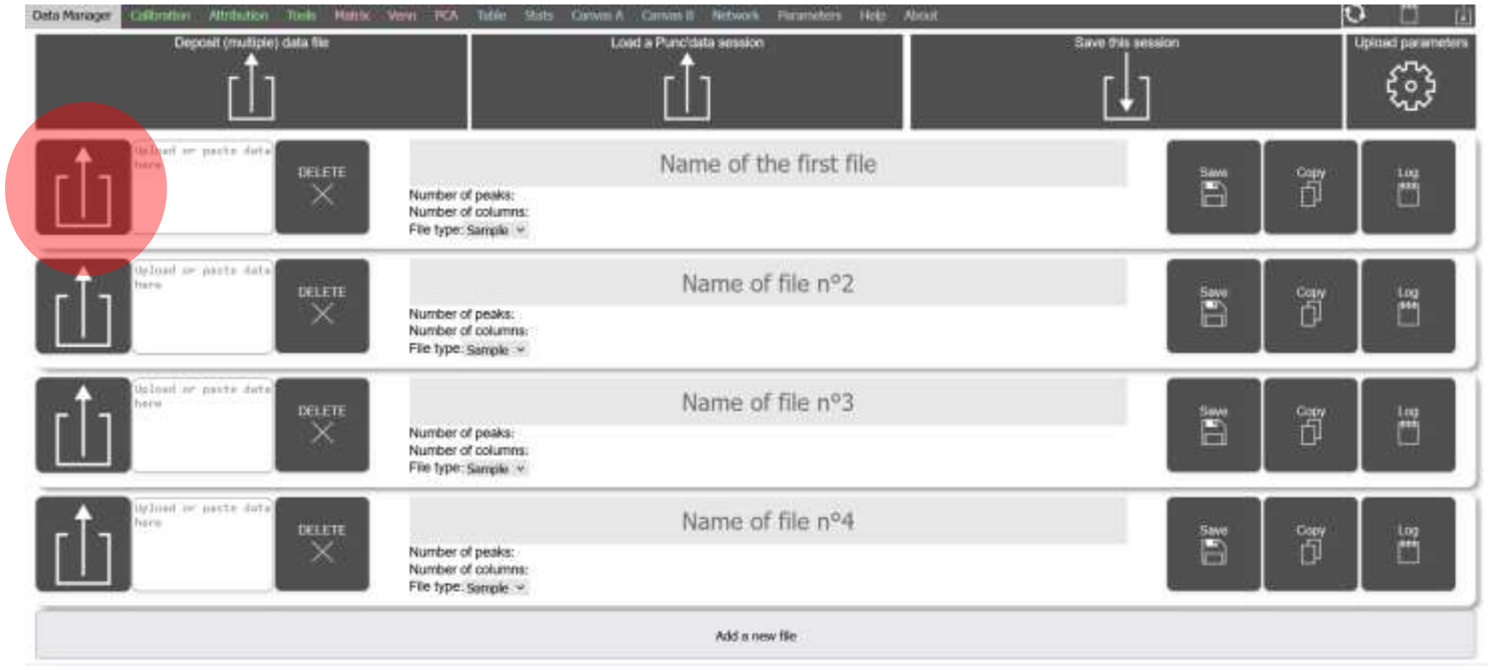
2.1. Opening Punc'data

- Start Punc'data, with a downloaded version or through wtvoe.github.io/puncdata/
- Tabs at the top help you navigate through the functions of Punc'data
- The ones you will need for this tutorial are highlighted below:

2.2 Importing data

Punc’data only accepts peaklists, not profile spectra. Lines should be for peaks, columns for data related to each peak
Any column order can be accepted.

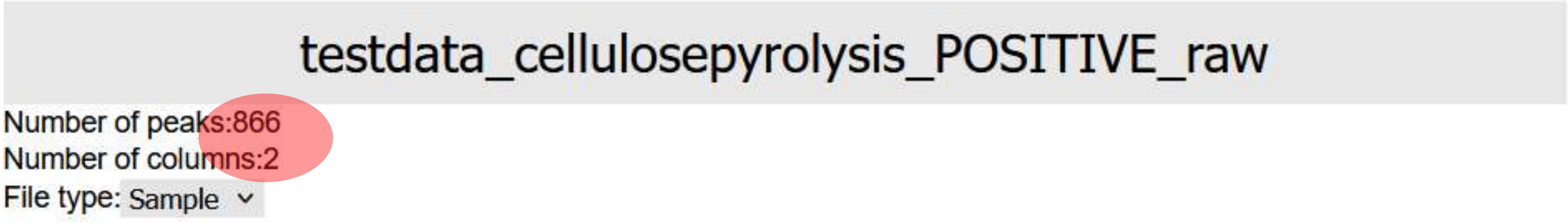
- Download the test data on Github : Examples\testdata_cellulosepyrolysis_POSITIVE_raw.csv



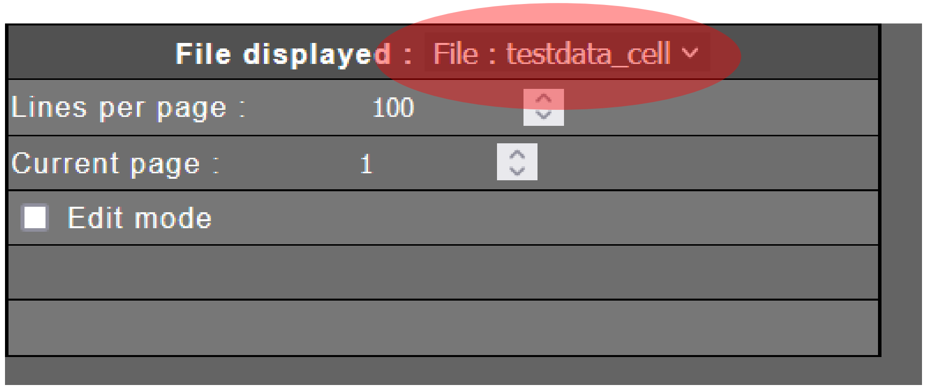
- Press the button highlighted in red and import your file
- *Alternatively you can paste it from a spreadhseet (CTRL+V) into the text area next to the highlighted button*
- You can then rename your file

2.3 Verifying that the import is correct

- A number of peaks and a number of columns should appear once the import is complete



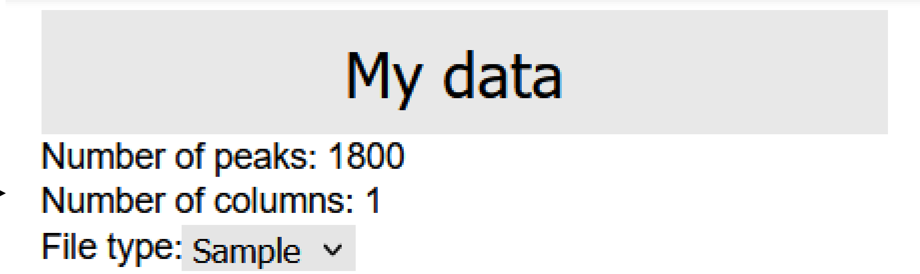
- Go to the tab « TABLE » and select your file in « file displayed » :
- Your file should appear below



For the test file: the columns are « m/z » and « I », with a line for each m/z value

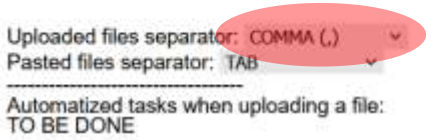
2.3 Common issue

- A common issue when importing data is related to the separator between columns in the uploaded file. When a problem occurs, on tab « DATA MANAGER » the number of columns will display only 1



- To fix this, press the button « upload parameters » on the top-right, with the gear icon

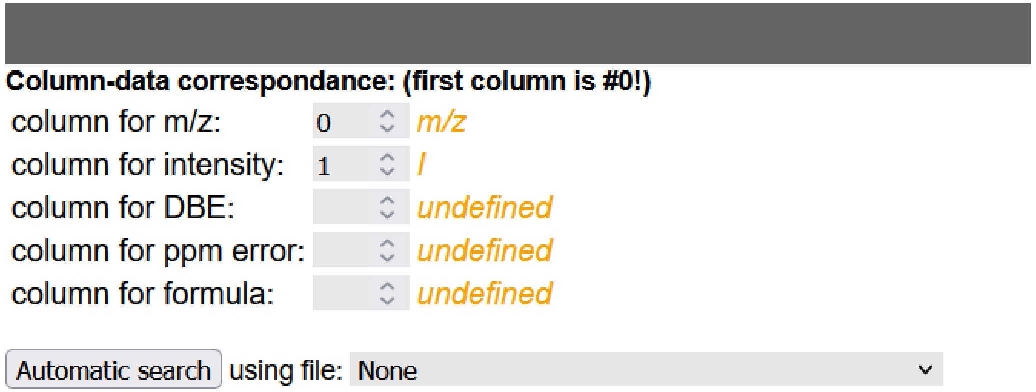
- A menu opens, select the correct option for your file in « uploaded file separator » and press save



- Press « DELETE » for your file and re-upload it.

2.3 Verifying that the import is correct

- Go to the tab « PARAMETERS »
- Punc’data should automatically identify your m/z and I columns, on the leftmost menu:



- If this is not correct for your data, you can set them by hand. Beware, the first column is numbered « 0 »

TIP: by selecting a file below (after « automatic search ») and pressing the button, the columns names in orange will be updated to show the correct names

3.1 Assignement: the basics

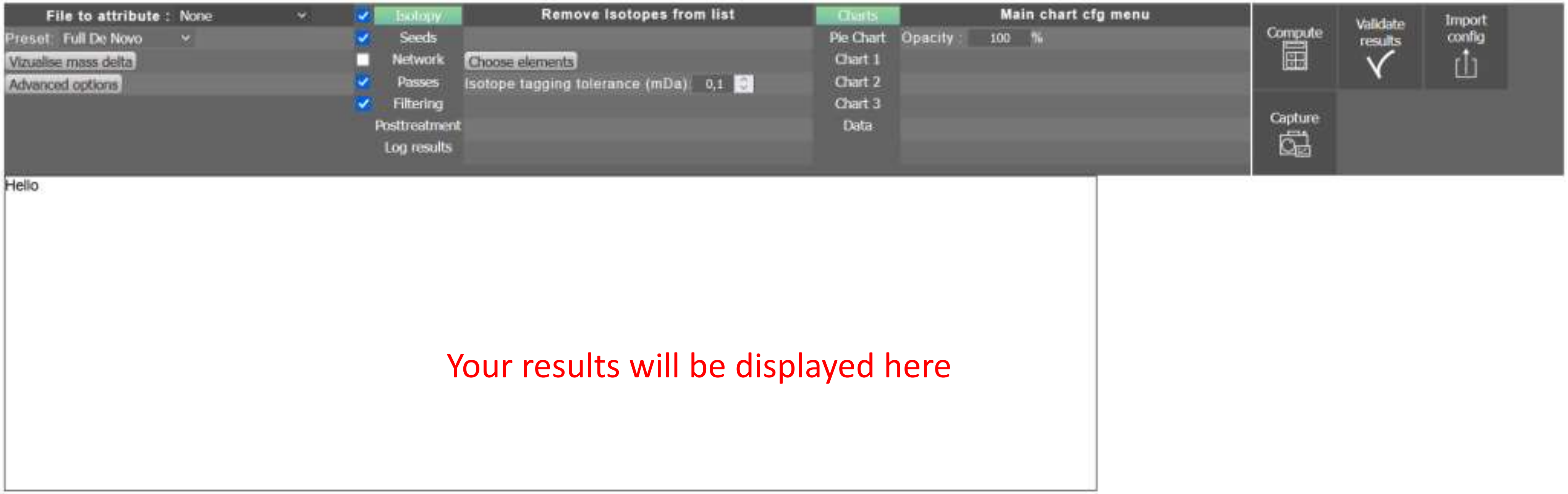
- Go to the third tab : « ATTRIBUTION »

Set up your parameters here

Assignment parameters

Display parameters

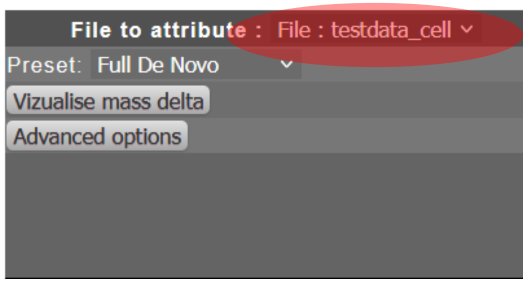
Start the process and save the results



Your results will be displayed here

3.1 Assignement: the basics

- Let’s start with the leftmost menu. Select your file in « file to attribute »



- For now, keep the preset « Full De Novo ». No mass differences will be computed

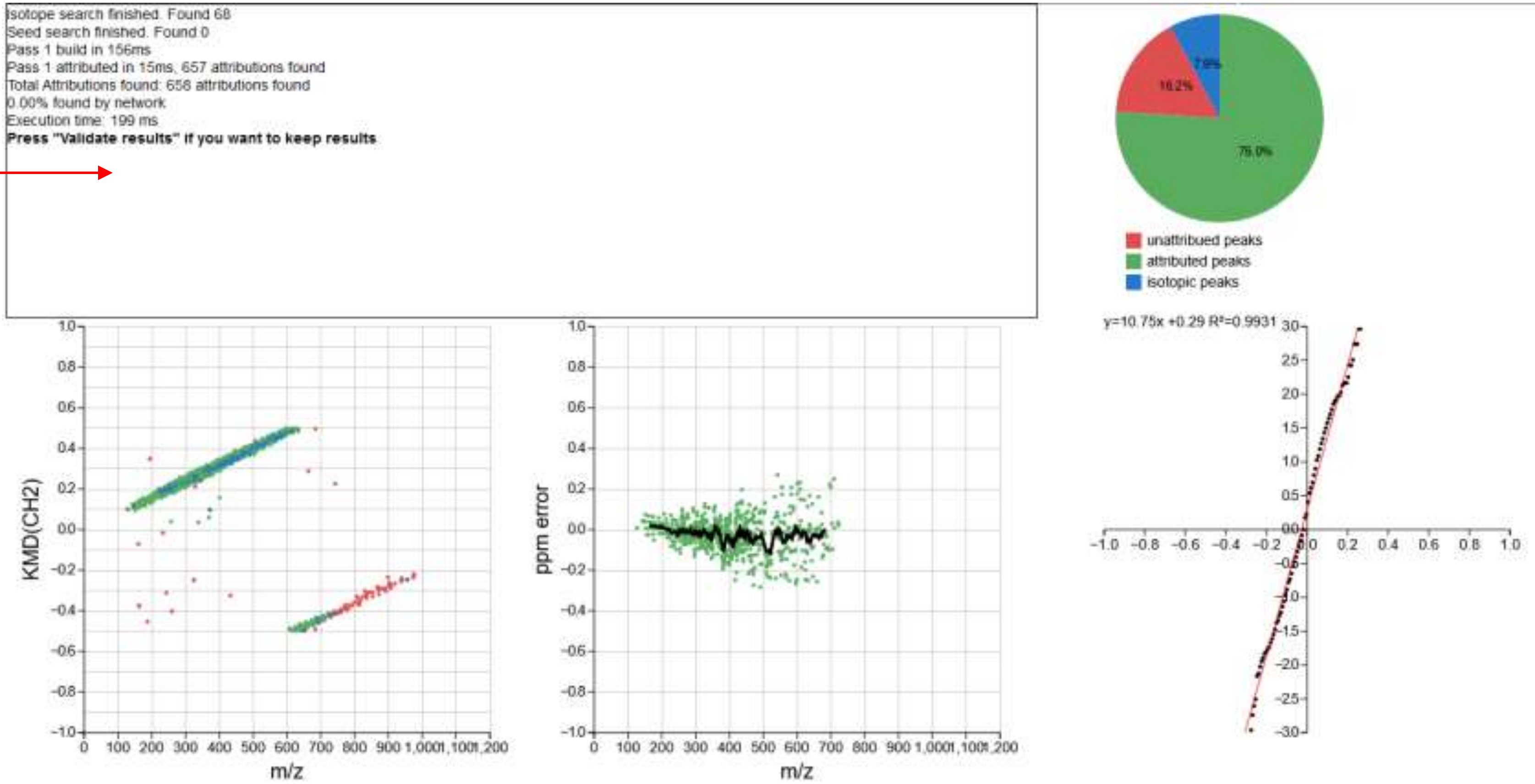
- Press « compute » on the rightmost menu

TIP: the menu « vizualise mass delta » will be a useful option later to display a specific mass difference histogram for your chosen sample

3.1 Assignement: the basics

- You should get these results
- The text box displays the progress of the algorithm

If it ever happens that the last line isn't displayed, it means that the algorithm encountered a problem and stopped

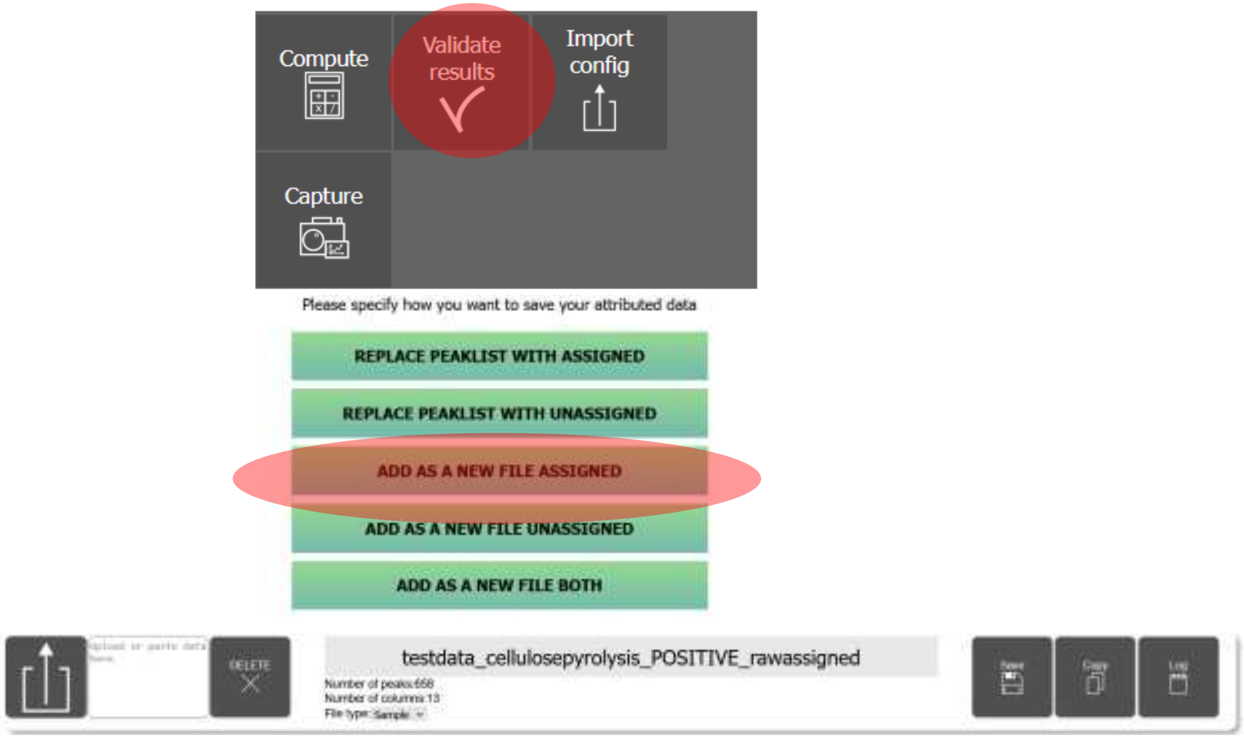


The pie chart represents the proportion of assigned peaks (with isotopes in blue) and non-assigned

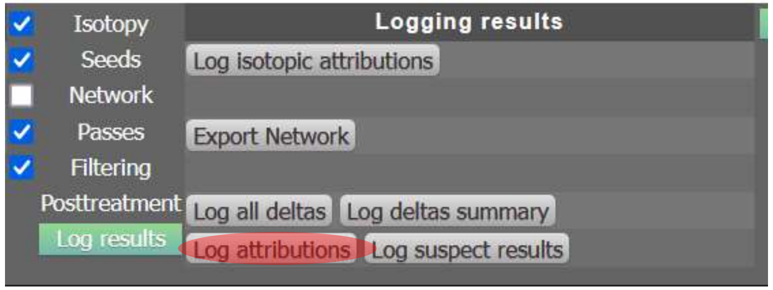
These charts can be modified, this will be mentioned in the last part of this tutorial (part 5)

3.1 Assignement: the basics

- Once you are happy with your results, press « Validate results »
- You will be asked what to do with the results. For this example, lets select « ADD AS A NEW FILE ASSIGNED »
- Go back to the « DATA MANAGER » tab. You results will appear as a new file, with 13 columns *(we will learn how to change this later*). You can then copy or save this dataset
- Alternatively, on the « ATTRIBUTION » tab, middle-left menu, you can press « Log results » and then « Log attributions ». The results table will be displayed

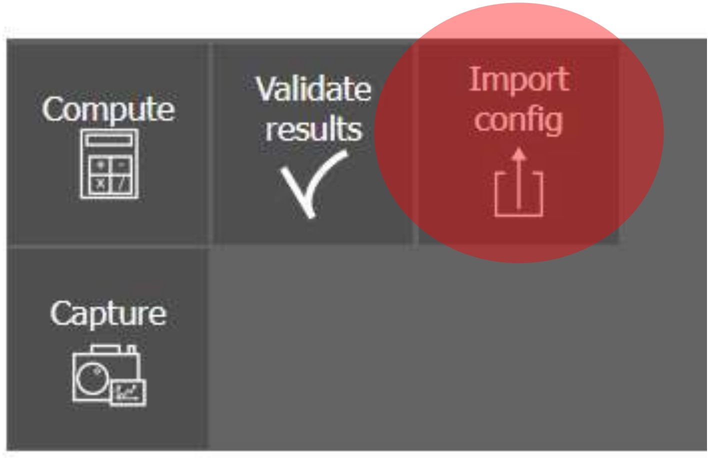


OR:

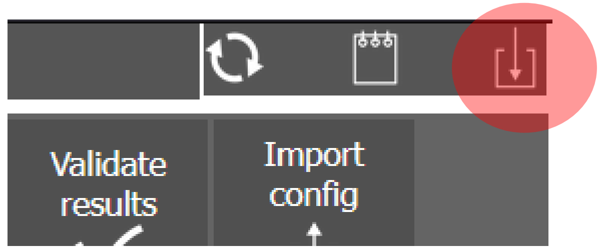


3.2 Import/export parameters

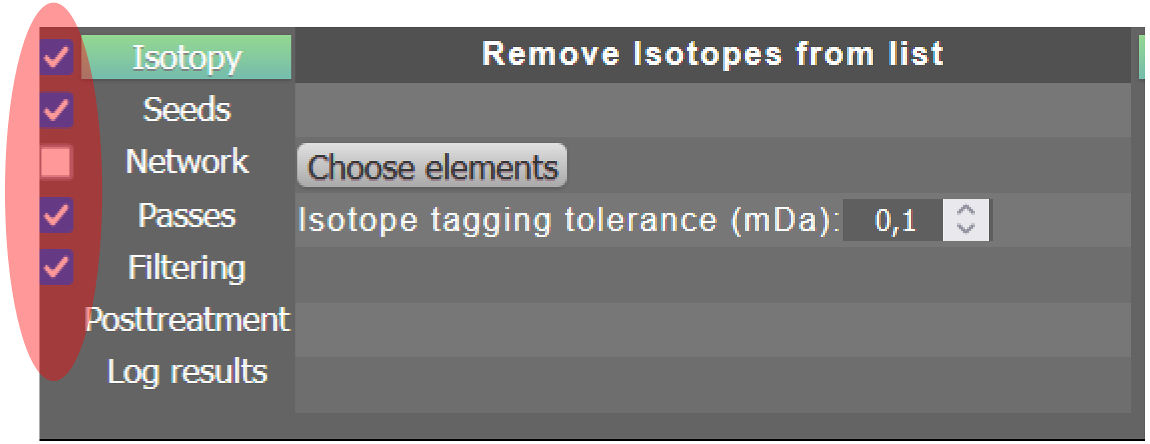
- If you already have customized parameters, you can import them by pressing « import config » and choosing your file (.json or .pdata)



- If you want to save your current parameters, press the top-right button next to the notebook. You can then export only the parameters (.json file) or the whole session, with datasets included (.pdata file)
- Alternatively, you can go to the « DATA MANAGER » tab to also save the session. The effect is identical.



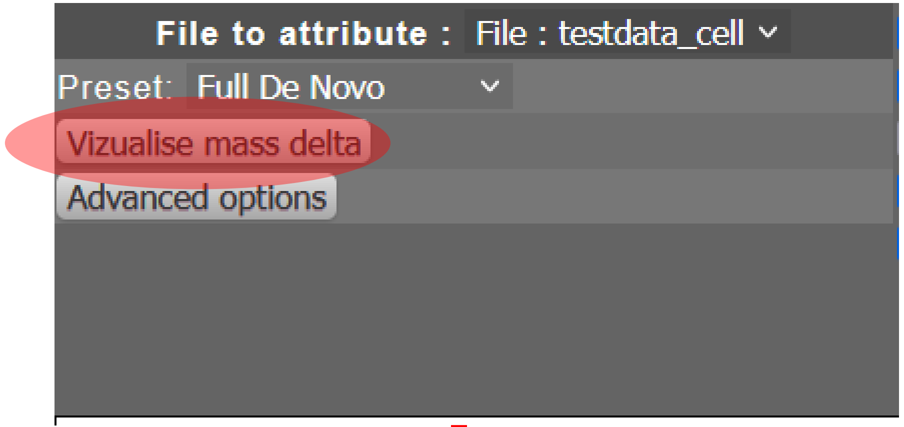
3.3 Modify parameters



- The assignement process is segmented in subprocesses. They can be checked in/out as you wish on the middle-left menu. You can try to check in/out some of them and see the effects on the assignment results after pressing « compute » again.
- « Isotopy » tags the isotopic distributions
- « Seeds » assigns a list of predefined ionic formulae
- « Network » is for creating a mass difference network that will assist assignments
- « Passes » is used to modify the De Novo assignment procedure
- « Filtering » is used to defined rules to filter out unrealistic formulae

Kind et al. BMC Bioinformatics 2007
https://doi.org/10.1186/1471-2105-8-105

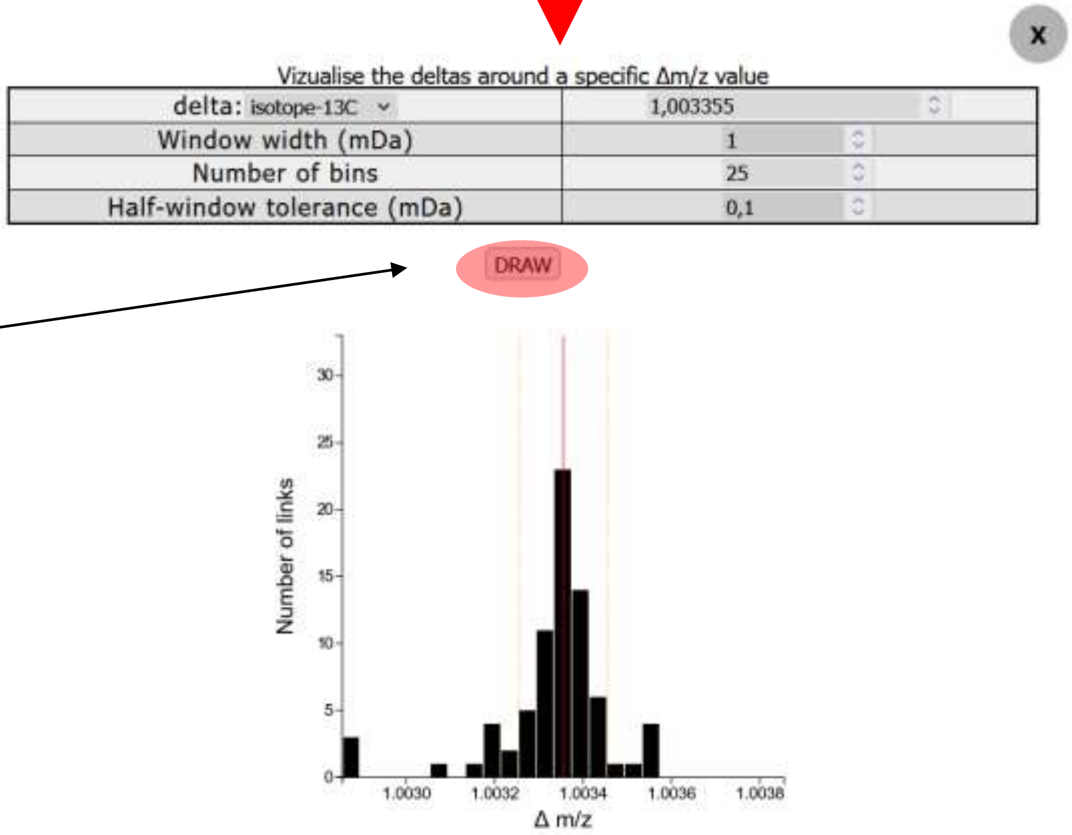
3.3.1 Isotopy



- « Isotopy » searches for isotopic mass differences and tags the peaks. Press « choose elements » to select which one you need. For the test sample, leave only ^{13}C.

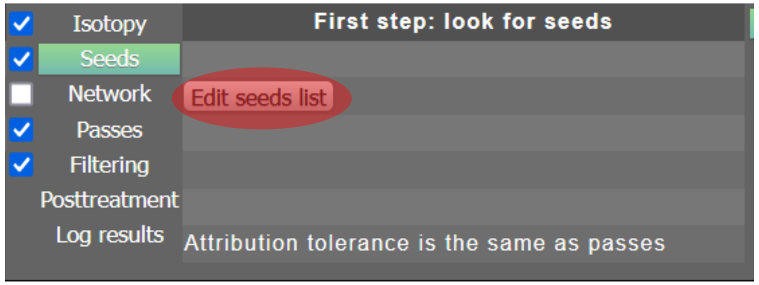
TIP: with « advanced edition » you can add other elements/isotopics distribution that could occur in your datasets and that you want to tag.

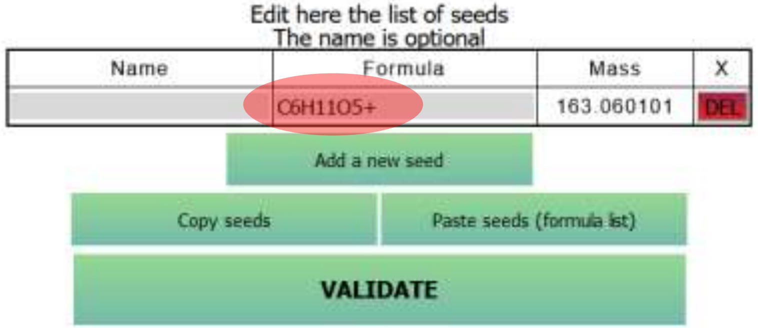
- The correct tagging tolerance depends on your sample. To estimate it, press « visualize mass delta » on the left menu and press « DRAW »
- Configure the width of the orange dotted lines with « Half-window tolerance »
- Once you find a value that encompasses your mass difference peak, use the same value for the isotope tagging tolerance. Here we will use 0.2 mDa



3.3.2 Seeds

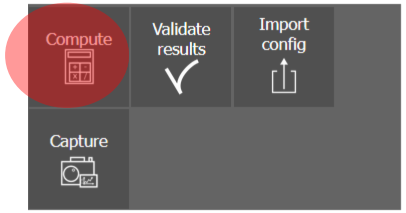
- You can add seeds, which will be the first formulae to be assigned. Press the button « Edit seeds list »
- You can paste a list of formulae from a spreadsheet (press paste seeds), OR you can write them one by one by pushing « Add a new seed »
- WARNING: you should use Ionic formulae, with a charge !

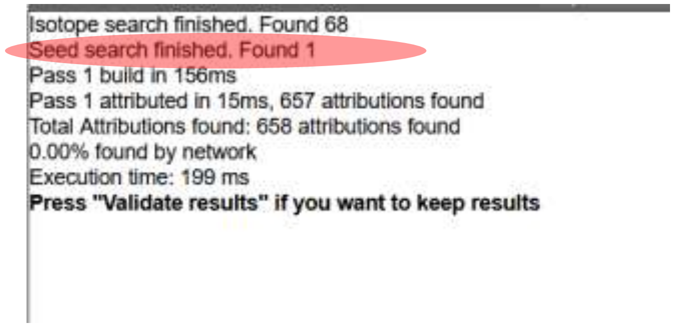




- For the example dataset, let's add **C6H11O5+** and press « VALIDATE »
- If you re-compute, the text box should now display:

« seed search finished. Found 1 »





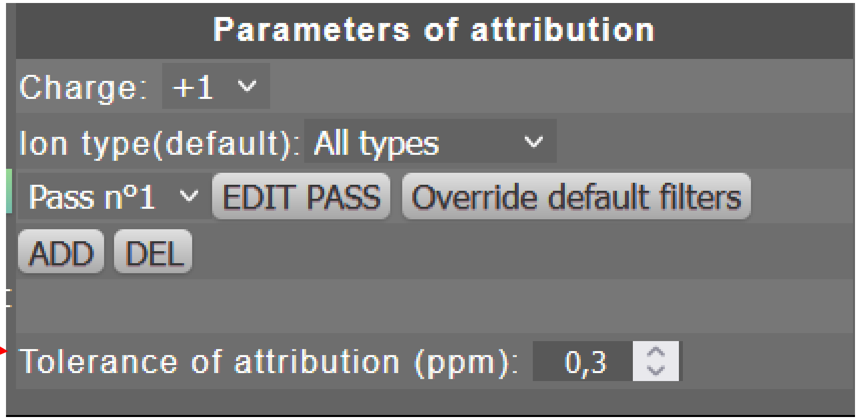
3.3.3 Networks

- The network submenu is a bit more complex. Go to **part 4** to better understand how to customize it best to your needs

- For now, keep the checkbox unchecked

3.3.4 Passes

- This menu helps you build (multiple) passes that will search the best element combination for every m/z value in your dataset within a defined tolerance



- For this sample, the ionization was done in APCI(+), so the correct charge is (+1) and the correct ion type is « All types ». *If you use ESI(-) for example, use charge (-1) and Ion Type « No radicals ».*

- *For this sample, you can leave the tolerance at 0.3 ppm*

The algorithm:

- Pass 1 formulae are computed
  - The best fit is searched for every *m/z* value in your dataset
    - For a defined m/z value, If at least one formula is found within the tolerance of attribution (ppm) and respects the filtering rules, the peak is assigned. If there are multiple formulae, the one with the lowest error is chosen
      - If you used a network, it is explored with this new assignment as a seed
- The algorithm then proceeds to Pass 2 with the non-assigned peaks only

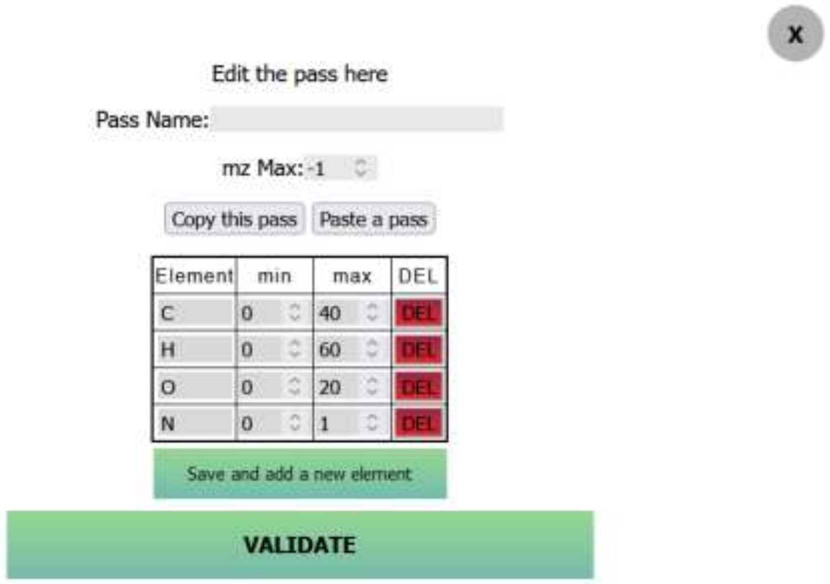
3.3.4 Passes

- Press « edit pass »
- A menu opens that allows you to customize the element list, the minimal and maximal value of each of them in any given formula

- Leave « mz Max » at -1 if you don’t want to define a maximal m/z value computed.
- You can copy/paste other passes from a spreadsheet. You can also name your pass if needed

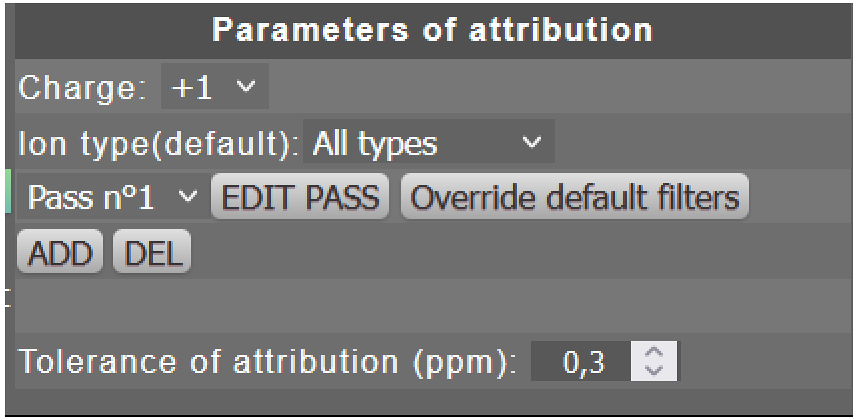
• When you press « VALIDATE » Punc’data alerts you if the pass exceeds 4 million possibilities, as computation times will get increasingly longer

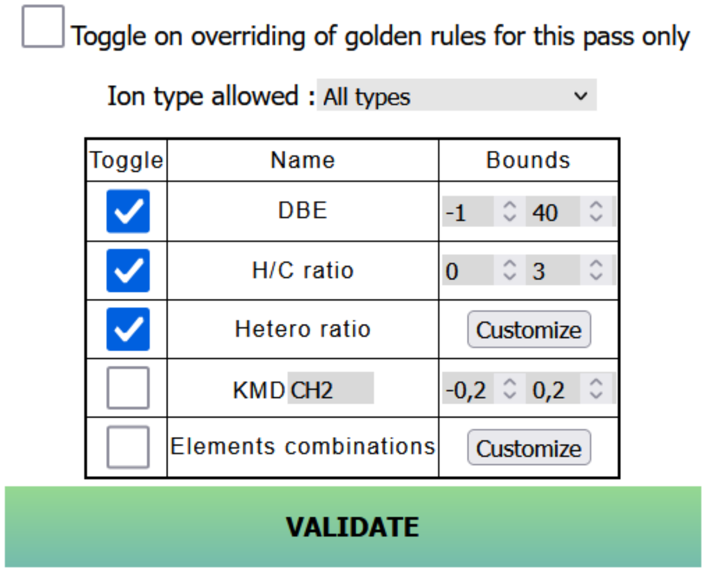
- For this dataset, the default pass is already good enough since we will use the network later.



3.3.4 Passes

- The button « ADD » creates a new pass. It will be empty at creation. Select it, and press « EDIT PASS » to modify it
- For the test dataset, only one pass will be enough
- The button « DEL » deletes the currently selected pass
- The button « Override default filters » will open a menu that is a copy of the « Filtering » menu, but which can be modified only for the currently selected pass
- If you want to try it, press « Toggle on overriding »
- We won't need it for our current example, as we only have one pass





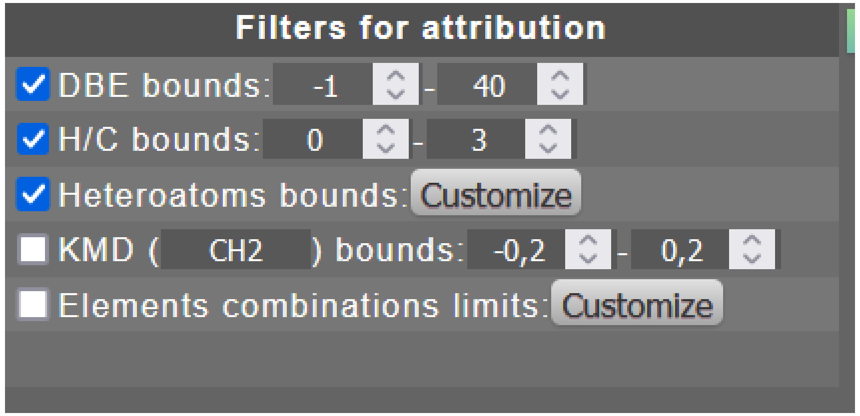
3.3.5 Filtering

Kind et al. BMC Bioinformatics 2007
https://doi.org/10.1186/1471-2105-8-105

- The default parameters of this menu are inspired by the « Seven Golden Rules »

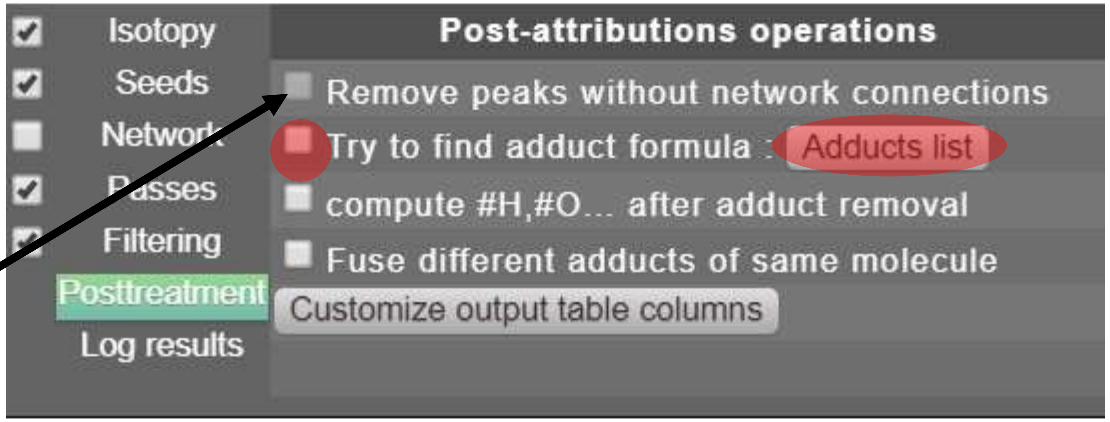
- These values are already good enough for a vast majority of datasets
- You can check in/out which rules are applied for every formula

- « KMD » (Kendrick mass defect) limits the assignement to only a subsection of KMD values between the bounds defined and for the repeat unit defined between the parenthesis

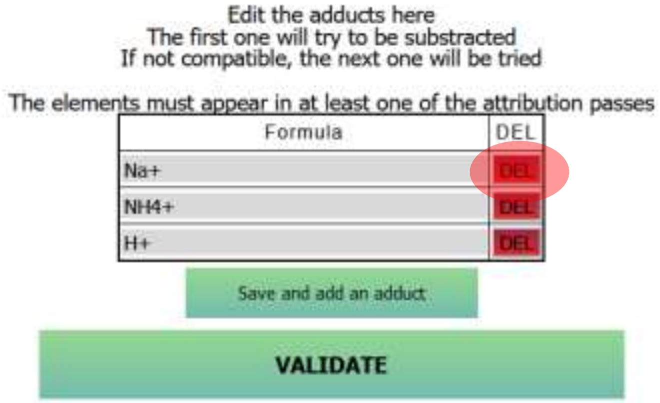


Kendrick E., Anal. Chem. 1963
DOI: 10.1021/ac60206a048

3.3.6 Post treatment



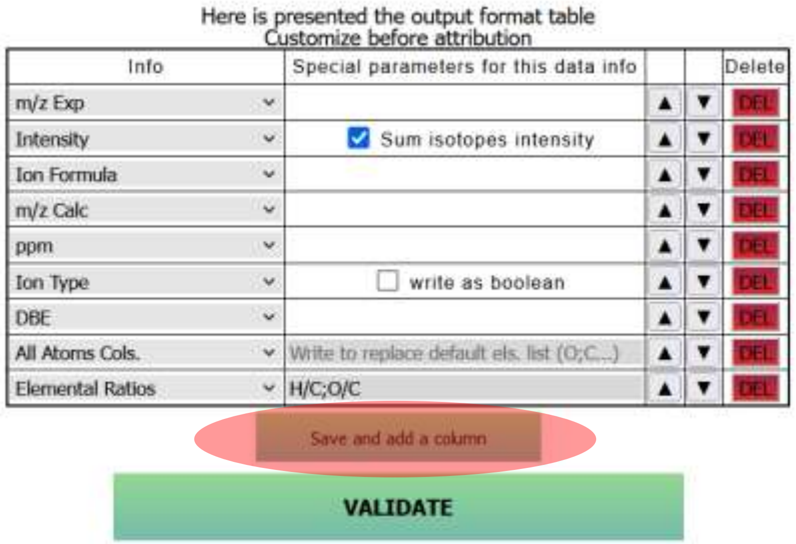
- « Posttreatment » include multiple options that can be toggled on/off after the assignement process is finished
- Peaks that do not have any network connection will be removed (it cannot be toggled on/off if you do not use a network)

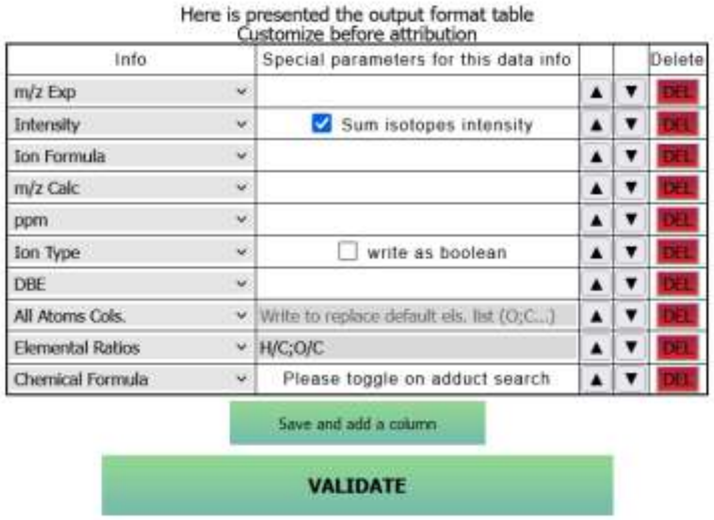


- The three options below allow you to try and compute « chemical formulae » instead of ionic formulae. Let's try it, check the box before « Try to find adduct formula » and then press « Adducts list »
- The algorithm will try to remove a Na^+ ion for each assignement, if it is not possible, it will try with $NH4^+$, and if it is still not possible (for example if there is no nitrogen in the ionic formula), then it will try to remove H^+
- For the test dataset you can remove « Na^+ » by pressing DEL and « VALIDATE »

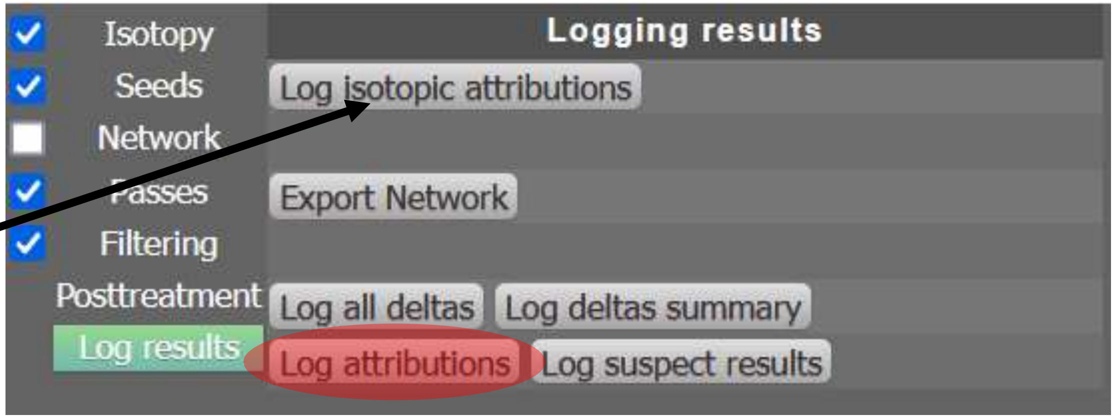
3.3.6 Post treatment

- Let’s then press « Customize output table columns ». This will allow you to modify the format of your results. You can use any column order, display additional data or even add the original columns of your peaklist, if you had for example the resolution of your peaks in a third column. Modify this as much as you need !
- For now, let’s try to add our « chemical formula » from the previous slide. Press « Save and add a column »
- A column named « empty » will appear. Change it to « Chemical Formula » and then press « VALIDATE »
- Re-compute the analysis





3.3.6 Log results



- In this menu you will be able to look at the tagged isotopes
- You also have many log options for the network and the mass differences. This can be useful when investigating the assignment process of a sample or the most common mass differences
- Here, let’s press « Log attributions ». The righmost menu should be « chemical formula », with the adduct and the charge removed (the new column we just added)

	ionType	DBE	#C	#H	#O	#N	H/C	O/C	Chemical Formula
9482	adduct	3.5	6	7	3	0	1.1666666666666667	0.5	C6H6O3
977	adduct	3.5	6	7	4	0	1.1666666666666667	0.6666666666666666	C6H6O4
4215	adduct	2.5	6	9	4	0	1.5	0.6666666666666666	C6H8O4
049	adduct	5.5	9	9	2	0	1	0.2222222222222222	C9H8O2
726	adduct	5.5	8	7	3	0	0.875	0.375	C8H6O3
119	adduct	4.5	9	11	2	0	1.2222222222222223	0.2222222222222222	C9H10O2
8728	adduct	4.5	8	9	3	0	1.125	0.375	C8H8O3
9456	adduct	3.5	7	9	4	0	1.2857142857142858	0.5714285714285714	C7H8O4
613	adduct	2.5	6	9	5	0	1.5	0.8333333333333334	C6H8O5
3962	adduct	6.5	10	9	2	0	0.9	0.2	C10H8O2
7053	adduct	6.5	9	7	3	0	0.7777777777777778	0.3333333333333333	C9H6O3
8335	adduct	1.5	6	11	5	0	1.8333333333333333	0.8333333333333334	C6H10O5
579	adduct	5.5	10	11	2	0	1.1	0.2	C10H10O2
3155	adduct	5.5	9	9	3	0	1	0.3333333333333333	C9H8O3
9824	adduct	4.5	9	11	3	0	1.2222222222222223	0.3333333333333333	C9H10O3

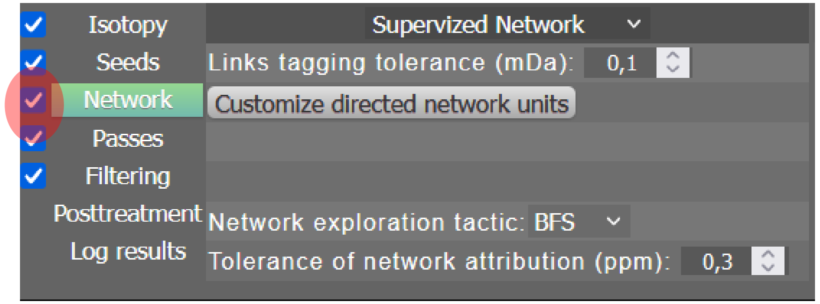
COPY TABLE

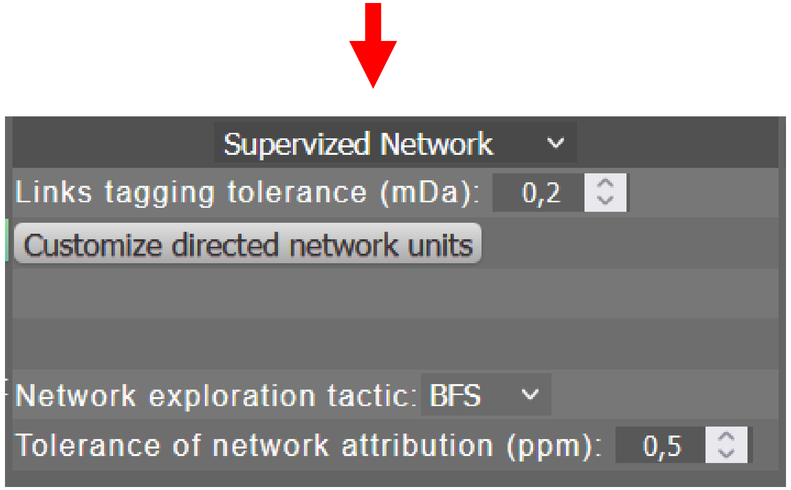
4. Network parameters

- There are two main network approaches. « Supervized network » where you select your own mass differences units, and « Unsupervized network » where the most common mass differences of your sample are searched and used to build the network.
- Choose « Supervized network » when you know the most common mass differences (CH_2, H_2…) and « Unsupervized network » for any other case (part 4.2 of this tutorial)

4.1 Supervized network

- First, check the « Network » box to toggle on supervized network
- For the links tagging tolerance, use 0.2 mDa (as a first approach for FT-ICR MS data, the same tolerance used for the isotopy is good enough)
- Do not change the network exploration tactic unless Punc'data notifies you of network errors. If there are, then changing it could be useful to verify that taking different paths in the network does not change the assignments made.
- For the tolerance, we can put a slightly higher tolerance than the *De Novo* one. Let's put 0.5 ppm





4.1 Supervized network

- Press the button « customize directed network units »
- A menu opens to change the units. The mass is automatically computed.
- We are analyzing cellulose. Let’s try to add the saccharide monomer, which is C6H10O5.
- Press « SAVE AND ADD A MASS », and write C6H10O5 in the « Formula » column
- Press « VALIDATE »

Edit the database here

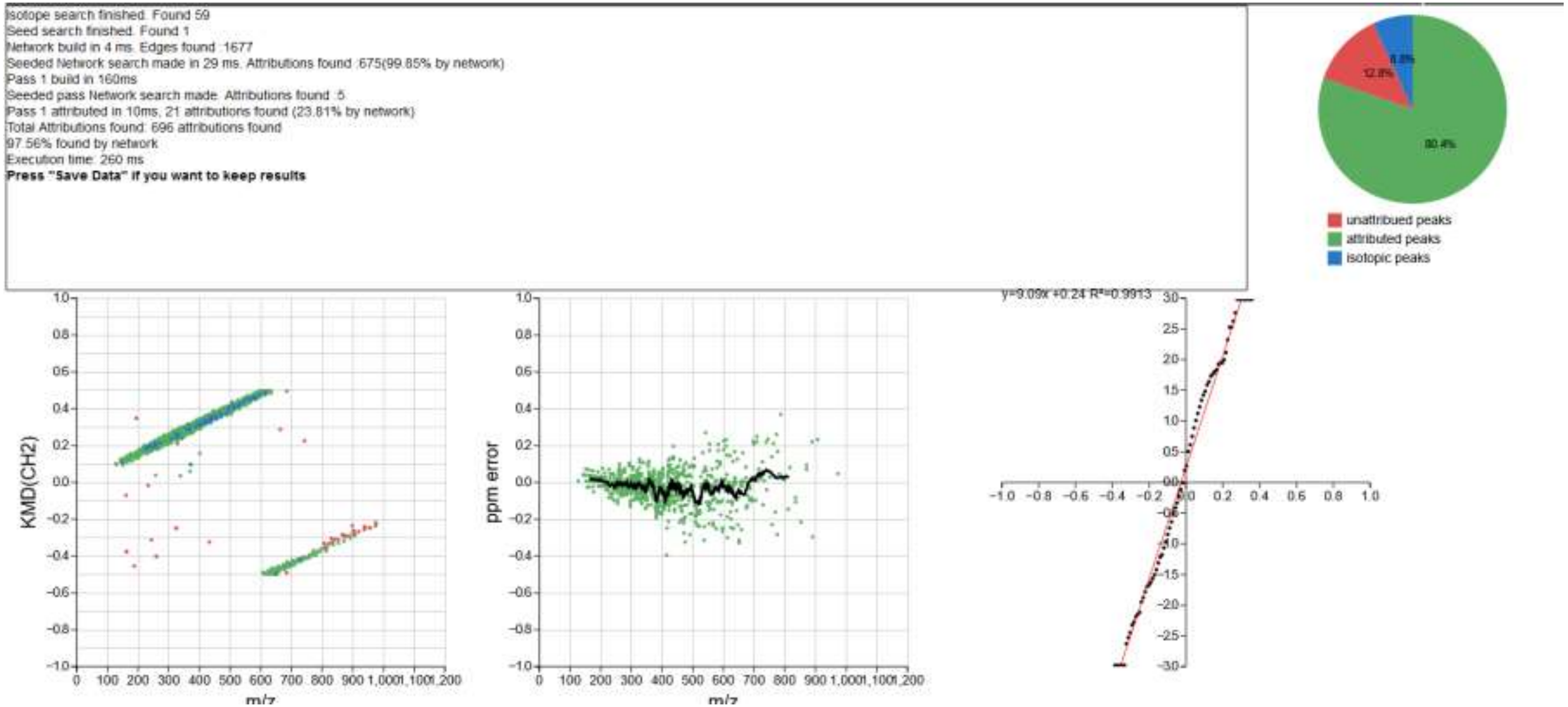
Formula	Mass	DEL
CH2	14,01565	DEL
H2	2,01565	DEL
H2O	18,010565	DEL
C6H10O5	162,052825	DEL

Save and add a mass

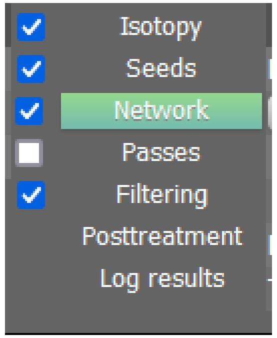
VALIDATE

4.1 Supervized network

- We customized our network. Press « COMPUTE ». New results should appear.



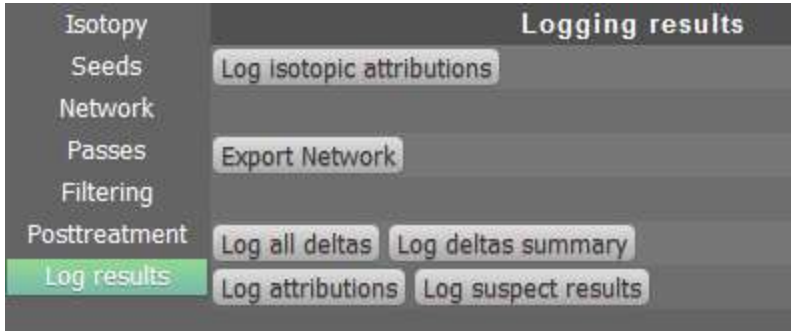
4.1 Supervized network



- We can now uncheck the « Passes » step, and then press « Compute »

- Slightly less assignments are found, but the algorithm is faster ! This can be quite noticeable on big datasets with >7000 peaks.
- You could still try and add more network units or more seeds to connect the lost assignements.

4.1 Supervized network



- If you want to look at your network results, press « Log results »

- « Log all deltas » will show you every mass difference found, with an indication of the two *m/z* values it connects

- « Log deltas summary » will display each mass difference type and the number of times it was found in the dataset

Data about the edges of the network:

Link type	# Occurences
CH2	420
H2	390
H2O	520
C6H10O5	347

COPY

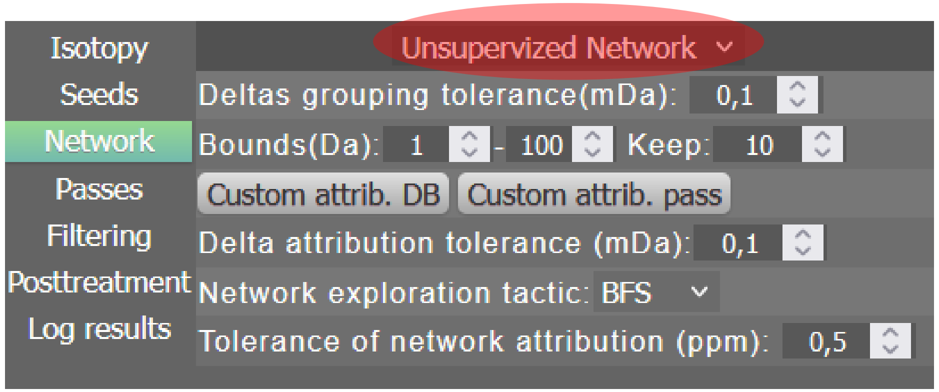
4.2 Unsupervized network

- On the top line, change the value to « Unsupervized Network »

- « Deltas grouping tolerance » defines how the algorithm groups together mass differences under the same Δ*m/z* value. Let's set it to the same value as your isotopy tagging tolerance (in this case, 0.2 mDa)

- The bounds define the range of mass differences that are searched. If you know that your sample may contain a polymer with a repeat unit >100 Da, increase the max value. Usually the most common mass difference have low Δ*m/z* value.1-100 is already good here.

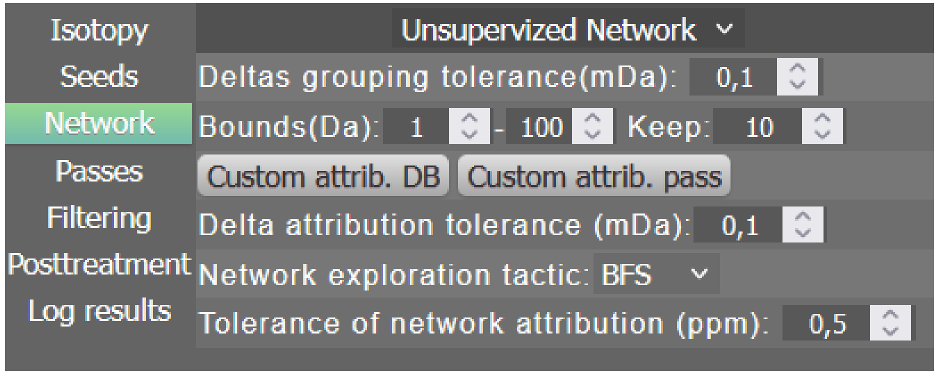
- « Keep » indicates how many of the most common mass differences are kept to build the network. 10 is already more than enough in most cases.



4.2 Unsupervized network

- Once the most common mass differences are found, Punc'data will try to assign them. It will look first through a database (customized through the button « custom attrib. DB ») and then through a small De Novo pass (customized through custom attrib. Pass »)
  For the test dataset, you will not need to modify anything

TIP: You can use negative number of elements in the custom attrib. pass



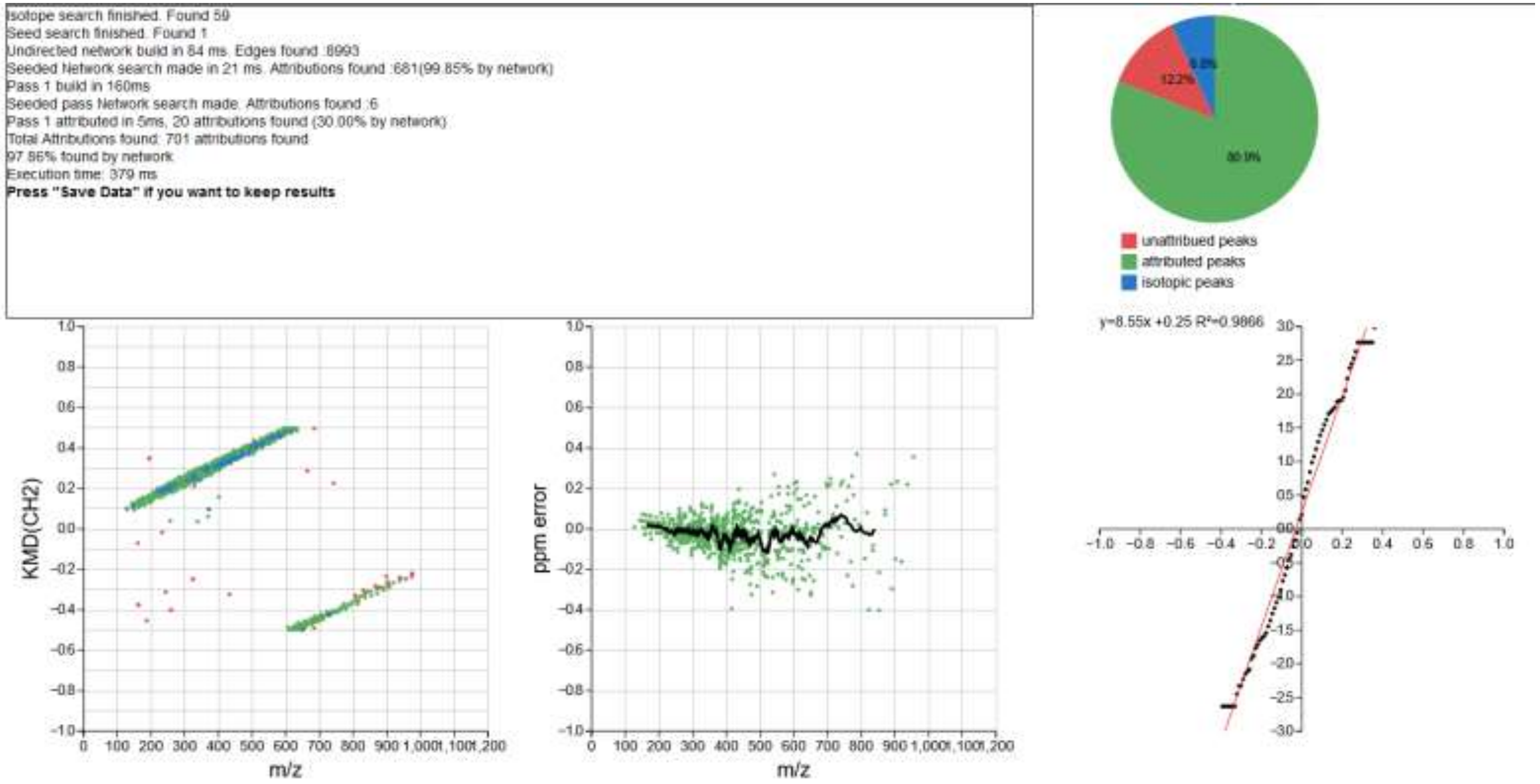
- The assignment tolerance of all mass differences is customized through the option « Delta attribution tolerance » . Keep it at 0.1 mDa.

Green and Perdue, Anal. Chem. 2015
DOI: https://doi.org/10.1021/ac504166t

- For the last two options, please refer to slide #27

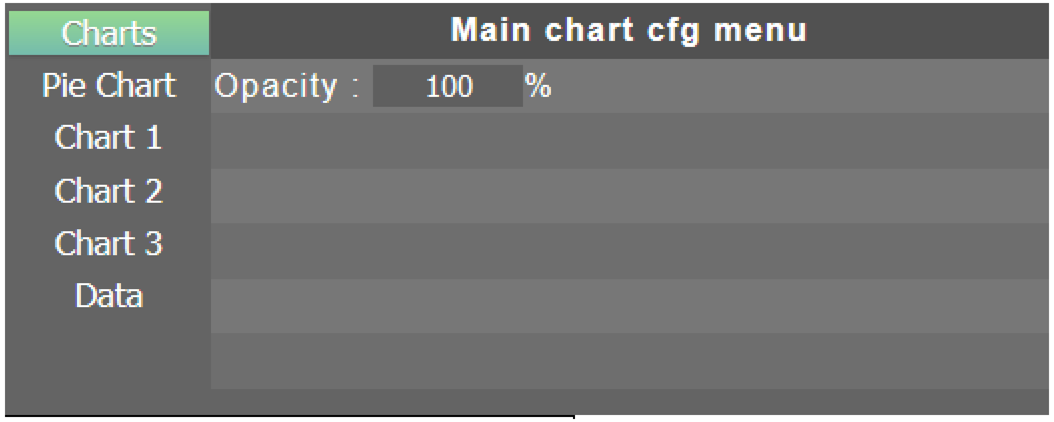
4.2 Unsupervized network

- Leave the parameters as is and press « Compute ». You should get more assignments



5. Customize assignment charts

- The charts on this tab can be customized with the middle-right menu
- « CHARTS » is currently only for dots opacity
- « PIE CHART » changes which sectors are displayed on the pie chart and if the sectors are relative to number of assignments or to total intensity
- « CHARTS 1/2/3 » can help you modify the three charts displayed below the text box
- « DATA » allows you to hide some subsets of data on all three charts

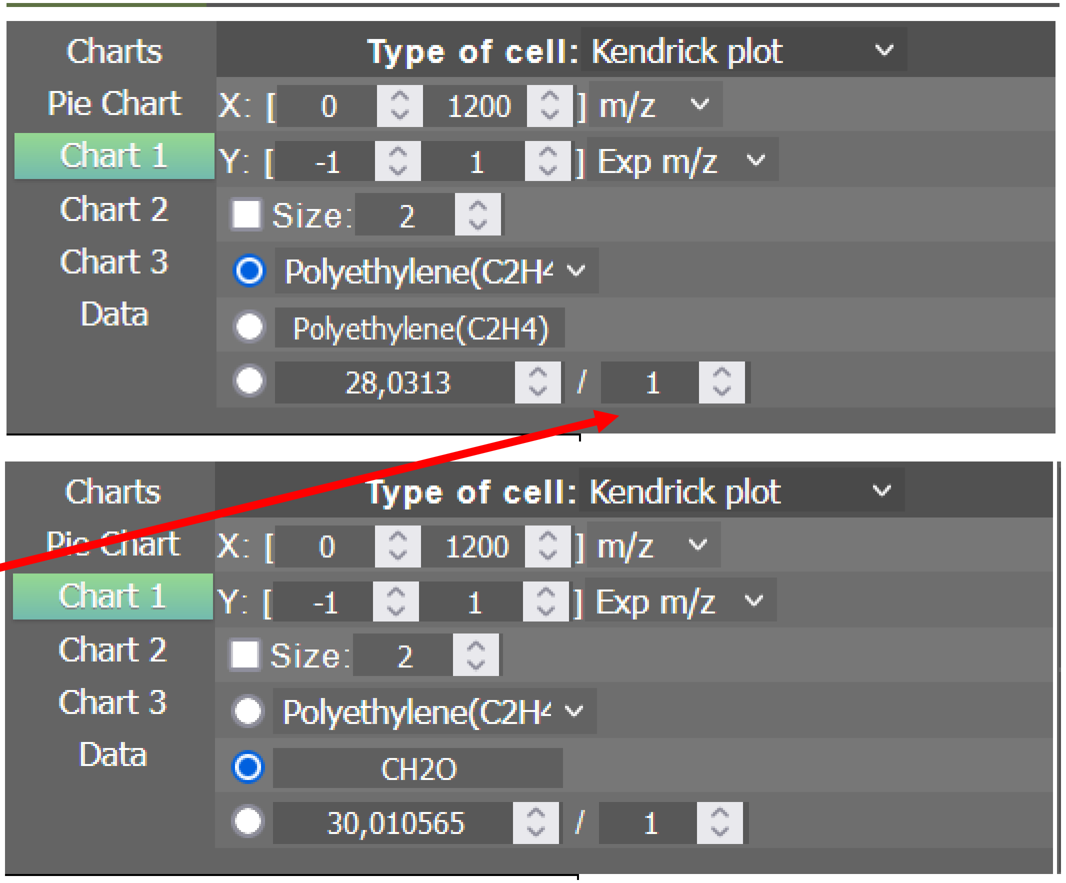


TIP: Many more chart type and functions/customisability exist in Punc'data, but are only available in « CANVAS » and « STATS »

5. Customize assignment charts

- Let's change Chart 1. Press « Chart 1 ». C2H4 is not adapted for cellulose. You can select another repeat unit in the menu on the 4th line, or press the button on the 5th line, write « CH_2O » and press enter.

- You can add a Kendrick divisor



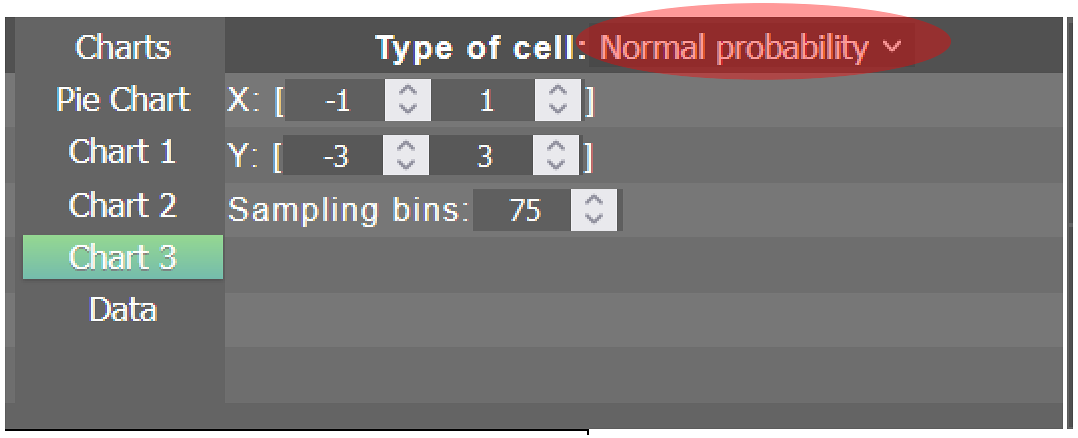
Fouquet et al. Mass Spectrometry 2017

DOI : 10.5702/massspectrometry.A0055

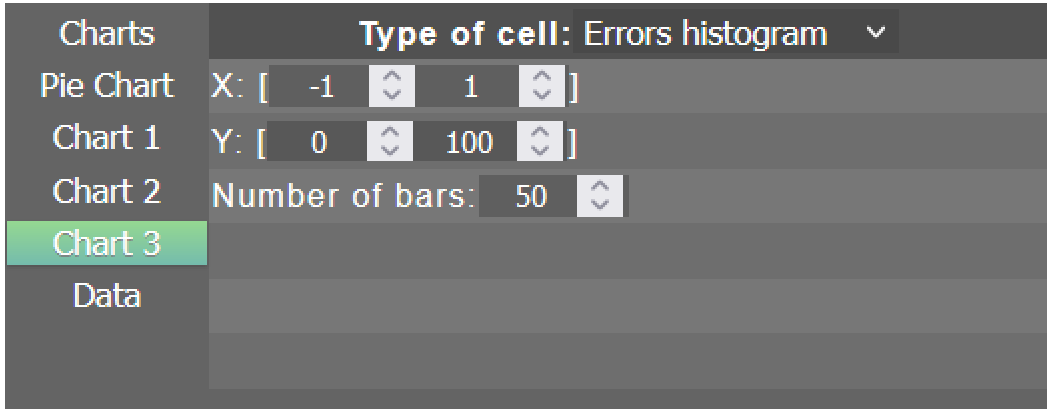
5. Customize assignment charts

- For Chart 3, let's change the normal probability plot to a histogram.

Press « Chart 3 », and on the top title line select « Errors histogram »



Put the Y values between 0 and 100, and « Number of bars » at 50. Press « COMPUTE » and you will now get your new chart.



You can press « CAPTURE » on the top-right menu to save as SVG or PNG any chart.

5. Customize assignment charts

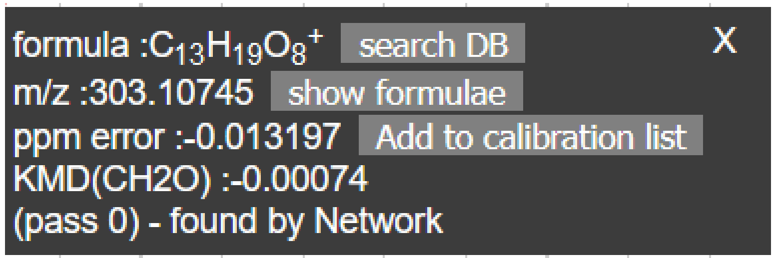
- Some Tips with charts:

- You can hover over points to display additional data on them: m/z ratio, assigned formula, ppm error, the pass with which they were assigned (Pass 0 for seeds) and if they were assigned as a seed, through passes or through network.

- You can select points to highlight them on other charts

- Shift+drag allows you to zoom on charts

- Ctrl+click makes a tooltip sticky. Additional buttons appear



« Show formulae » shows you alternative formulae for this m/z ratio with a custom pass assignment.

Do you have any questions ?

- Please adress any comments, suggestions or questions at :

- lcpa2mc-puncdata-contact@protonmail.com
- Or through GitHub : https://github.com/WTVoe/puncdata/discussions